

- Cyber security digital twin for software-defined aircraft.

⇒ Why this project?

Modern aircrafts are becoming software-defined systems.

Not just engines and wing. They contain:

All these systems continuously exchange data between them for functioning.

- Navigation software
- Flight management system.
- Communication systems.
- Sensor networks.
- Avionics networks.

Now, attackers may try to feed false information into systems that rely on data and cause disruption.

(i) GPS spoofing

(ii) Sensor spoofing

(iii) Communication data manipulation.

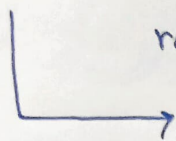
(iv) Network intrusion.

Objective: If false data starts appearing, can AI recognize that something is wrong?

Proposed solution: • AI continuously reads the incoming data and detect anomalies present and flags it.

- The system examines the anomaly.
- Use digital twin.

Digital twin: Simply a software representation of a real system.



Acts as a virtual aircraft model.

Used to simulate how threats and mitigation actions affect those systems.

- (i) Which part of aircraft affected?
- (ii) How does the threat spread?
- (iii) What happens if we take action?



Using digital twin, we can come up with:

- Impact simulations.
- Mitigation recommendations.
- Mitigation simulation (Apply the recommendations and recalculate).

Uses of this project:

(i) Early threat detection.

(ii) Aircraft system visibility.

(iii) Cyber Security training & simulation.

(iv) Decision support for aircraft threats.

(v) Impact assessments

(vi) Testing mitigation strategies.

(vii) MRO (Maintenance, Repair, Overall) Training.